

# CLEAR-Papaya: A Leakage-Audited, Calibration-Aware and Edge-Deployable Framework for Papaya Leaf Disease Recognition under Field-Realistic Distribution Shift

Mursalin Hawlader (Shuvo)<sup>a,\*</sup>, Md. Shovon<sup>a</sup>, [Third Author]<sup>b</sup>

<sup>a</sup>Department of Computer Science and Engineering, Netrokona University, Netrokona, Bangladesh

<sup>b</sup>[Affiliation — e.g. Department of Plant Pathology / Horticulture, collaborating institution]

\*Corresponding author: [email] ORCID: [0000-0000-0000-0000]

## ABSTRACT

**Background.** Papaya (*Carica papaya* L.) is a high-value tropical fruit crop whose yield is repeatedly devastated by foliar and systemic diseases — most notably Papaya Ringspot Virus (PRSV), anthracnose, powdery mildew and bacterial leaf spot. Deep learning promises smartphone-based early diagnosis for smallholder growers, and the published literature reports accuracies routinely exceeding 99%.

**Problem.** We show that this figure is largely an artefact of methodology rather than a measure of capability. Existing papaya leaf disease studies (i) split near-duplicate images of the *same physical leaf* across train and test partitions, (ii) evaluate exclusively on single-source, uniform-background laboratory imagery, (iii) never test whether the classifier is reading the lesion or the background, (iv) report no calibration, no open-set behaviour and no statistical testing, and (v) claim edge deployability without measuring it. **Method.** We introduce **CLEAR-Papaya**, a framework with two coupled components. First, **PLD-Bench**, an evaluation protocol built on a perceptual-hash and embedding-based *leakage audit*, group-wise splitting at leaf-instance granularity, a background-ablation *Clever-Hans probe*, a corruption suite, a cross-source shift split, and an open-set split containing diseases never seen in training. Second, **PapayaFormer**, a hybrid convolution–transformer classifier that couples a lightweight ConvNeXt/MobileViT backbone with a Multi-Scale Lesion Attention (MSLA) module and an evidential Dirichlet head, trained with a class-balanced focal–evidential objective and quantitatively verified explanations. **Results.** – (to be completed after experiments; all result tables in this manuscript are pre-registered scaffolds with the exact statistics, tests and confidence intervals to be reported.) **Conclusion.** –

**Keywords:** papaya leaf disease; *Carica papaya*; precision agriculture; data leakage; near-duplicate detection; distribution shift; uncertainty calibration; evidential deep learning; open-set recognition; explainable AI; edge inference; reproducible benchmarking.

## HIGHLIGHTS

- A leakage audit at leaf-instance granularity replaces random splitting and quantifies how much reported papaya-disease accuracy is inflated.
- A background-ablation probe tests directly whether the network classifies the lesion or the acquisition setting.
- PapayaFormer adds multi-scale lesion attention and an evidential head giving calibrated confidence and open-set rejection of unseen diseases.
- Explanations are validated quantitatively (deletion/insertion AUC, pointing game against expert lesion masks), not shown as cherry-picked heatmaps.
- Edge feasibility is measured, not asserted: INT8 latency, memory and energy on commodity Android and Raspberry Pi hardware.
- All splits, hashes, seeds, configs and code are released; every claim is accompanied by a statistical test and a bootstrap confidence interval.

### How to read this document

This manuscript is a *complete structural prototype*. Prose, formulations, protocols, algorithms, tables, figure slots and back matter are written in submission-ready form. Cells marked – are numeric results that must come from your own experiments — they are deliberately left empty rather than invented. Entries marked † are external resources (dataset DOIs, prior-work numbers) that you must verify at the source before submission. Section [Appendix G](#) gives the execution plan that fills every –.

## NOMENCLATURE

| Symbol / Abbr. | Meaning                                                       |
|----------------|---------------------------------------------------------------|
| $\mathcal{D}$  | Labelled image dataset $\{(\mathbf{x}_i, y_i, g_i)\}_{i=1}^N$ |
| $g_i$          | Group (leaf-instance) identifier of sample $i$                |
| $K$            | Number of disease classes                                     |
| $\alpha$       | Dirichlet concentration parameters (evidential head)          |
| $u$            | Predictive uncertainty mass, $u = K/S$                        |
| $\rho$         | Leakage rate of a split                                       |
| $\tau$         | Open-set rejection threshold                                  |
| MSLA           | Multi-Scale Lesion Attention module                           |
| EDL            | Evidential Deep Learning                                      |
| ECE / MCE      | Expected / Maximum Calibration Error                          |
| MCC            | Matthews Correlation Coefficient                              |
| AURC           | Area Under the Risk–Coverage curve                            |
| FPR@95         | False positive rate at 95% true positive rate                 |
| PRSV           | Papaya Ringspot Virus                                         |
| OOD            | Out-of-distribution                                           |
| GSKF           | Group-Stratified $k$ -Fold                                    |

## 1. INTRODUCTION

### 1.1. Agronomic motivation

Papaya (*Carica papaya* L.) is cultivated across more than 60 tropical and subtropical countries and is one of the fastest-maturing fruit crops available to smallholders, reaching commercial bearing within a single year of transplanting †. That short cycle is also its vulnerability: a single systemic infection can propagate through an orchard before the first harvest is realised. Papaya Ringspot Virus (PRSV) alone has repeatedly rendered commercial papaya cultivation non-viable in entire

districts, and foliar pathogens such as *Colletotrichum gloeosporioides* (anthracnose), *Oidium caricae* (powdery mildew) and bacterial leaf-spot complexes cause chronic defoliation, reduced photosynthetic area and quality downgrading<sup>†</sup>.

The economically decisive property of all four is that their earliest reliable signature appears *on the leaf*: chlorotic mottling and vein-clearing for PRSV, water-soaked necrotic margins for anthracnose, white superficial mycelium for powdery mildew, angular translucent lesions for bacterial spot. A grower who can distinguish these four patterns within days of onset can act; a grower who waits for fruit symptoms cannot. Extension pathology services, however, are thin in exactly the regions where papaya is grown intensively, and diagnosis latency is measured in weeks.

## 1.2. Why another deep learning paper is not obviously needed — and why this one is

A smartphone camera plus a convolutional network is the obvious response, and the literature has responded enthusiastically. Papaya-specific and general plant-disease studies report test accuracies of 96–99.9% with striking consistency<sup>†</sup>. If those numbers were transferable, the problem would be solved.

They are not, and the reason is methodological rather than architectural. Plant-disease image collections are typically assembled by photographing a modest number of physical leaves many times — multiple angles, multiple crops, multiple burst frames — under controlled illumination against a uniform background. A uniformly random train/test split then places *photographs of the same leaf, minutes apart* on both sides of the partition. The resulting metric measures near-duplicate retrieval, not disease recognition. Compounding this, a background that co-varies with class (different collection sessions for different diseases) offers a shortcut that gradient descent will take: the network can achieve high accuracy without ever attending to a lesion.

Neither failure is hypothetical, and neither is detected by any evaluation currently standard in this literature. Both are trivially detectable once one looks — which is what this paper does.

## 1.3. Research gaps

We synthesise seven concrete gaps (G1–G7) from the papaya and broader plant-pathology vision literature; Table 2 maps each to the contribution that closes it.

### G1 — Unaudited near-duplicate leakage.

No published papaya leaf disease study reports a duplicate or near-duplicate audit, or splits by leaf instance. Reported accuracy is therefore an upper bound of unknown tightness.

### G2 — Background confounding is never tested.

No study ablates the background to verify that the decision is lesion-driven.

### G3 — Single-source evaluation.

Models are trained and tested on one collection; cross-source and field transfer are unmeasured.

### G4 — Closed-world assumption.

Every image is forced into a known class. A farmer photographing nutrient deficiency, herbicide burn or an

unmodelled pathogen receives a confident, wrong diagnosis.

### G5 — Uncalibrated confidence.

Softmax scores are reported as "confidence" without reliability diagrams, ECE or risk–coverage analysis, although the intended use is triage under human review.

### G6 — Decorative explainability.

Grad-CAM figures are presented qualitatively on selected successes, with no faithfulness metric and no comparison against expert-annotated lesion regions.

### G7 — Asserted rather than measured deployability.

"Lightweight" and "suitable for mobile deployment" are claimed from parameter counts, without quantised latency, memory or energy measurements on target hardware, and without statistical testing across seeds.

## 1.4. Contributions

### C1 — PLD-Bench protocol.

A reproducible evaluation protocol for papaya leaf disease comprising a two-stage leakage audit (perceptual hash → self-supervised embedding clustering → union-find grouping), group-stratified splitting, and a published leakage rate  $\rho$  for every split (Section 3.5).

### C2 — Clever-Hans probe.

A three-way background ablation (full image / leaf-only / background-only) that quantifies the fraction of accuracy attributable to non-lesion context, reported as a mandatory dataset-quality statistic (Section 3.6).

### C3 — PapayaFormer.

A hybrid CNN–transformer classifier with a Multi-Scale Lesion Attention module targeted at the mixed-scale nature of papaya symptoms (diffuse mottling vs. punctate necrosis), and an evidential Dirichlet head enabling calibrated confidence and threshold-based rejection in a single forward pass (Section 4.2).

### C4 — Open-set and shift evaluation.

Explicit splits for unseen-disease rejection, cross-source transfer and a 15-way corruption suite at 5 severities (Section 3.7).

### C5 — Quantitatively verified explanations.

Deletion/insertion AUC, pointing game against expert lesion masks, and attribution stability under input perturbation, compared across four attribution methods (Section 5.4).

### C6 — Measured edge deployment.

FP32/FP16/INT8 latency, peak RAM, model size and energy per inference on a mid-range Android device and a Raspberry Pi 4, with the accuracy cost of quantisation reported per class (Section 5.5).

### C7 — Full reproducibility artefact.

Split manifests with per-image hashes, seeds, environment lockfiles, training configs, checkpoints and evaluation scripts released under a permissive licence (Appendix A).

## 1.5. Research questions

**RQ1:** How do standard backbones compare on papaya leaf disease under a leakage-free protocol? **RQ2:** How large is the accuracy inflation caused by random (leaky) splitting? **RQ3:** How much of the measured accuracy survives background ablation? **RQ4:** How does performance degrade under cross-source shift and synthetic corruption? **RQ5:** Can the model reject unseen diseases, and at what cost? **RQ6:** Is the reported confidence calibrated? **RQ7:** Which architectural components matter (ablation)? **RQ8:** Are the explanations faithful and lesion-aligned? **RQ9:** Is the model deployable on commodity edge hardware? **RQ10:** How does performance scale in the low-data regime relevant to new geographies?

## 1.6. Organisation

Section 2 reviews prior work and formalises the gap. Section 3 describes data sources, the curation and leakage-audit protocol, and the split design. Section 4 presents PapayaFormer and the training objective. Section 5 specifies the experimental protocol, metrics and statistical procedure. Section 6 reports results. Section 7 interprets them, Section 8 states threats to validity and limitations, and Section 9 concludes. Appendices provide the reproducibility checklist, the full hyperparameter grid, per-class results and a pre-emptive reviewer response.

## 2. RELATED WORK

### 2.1. Classical feature engineering

The pre-deep-learning line of work extracted colour statistics (HSV/Lab histograms, colour moments), texture descriptors (GLCM contrast/homogeneity/energy, LBP, Gabor banks) and shape features from segmented lesion regions, and classified them with SVM,  $k$ -NN or Random Forest<sup>†</sup>. These pipelines are interpretable and data-efficient, and remain a legitimate baseline — we include one (Section 4.6) precisely because a modern architecture that cannot beat GLCM+SVM by a statistically significant margin has not been shown to be necessary. Their weakness is segmentation: threshold- and clustering-based lesion segmentation degrades sharply outside controlled illumination.

### 2.2. CNN transfer learning

The dominant paradigm fine-tunes an ImageNet-pretrained backbone (VGG16/19, ResNet-50/101, InceptionV3, DenseNet-121, Xception, EfficientNet-B0–B7, MobileNetV2/V3) on plant-disease imagery [1, 2]<sup>†</sup>. Mohanty et al. [1] established both the effectiveness of the approach and — explicitly, in their own discussion — its central caveat: accuracy collapsed when the model was applied to images taken under conditions different from the training set. That caveat has been widely cited and rarely acted upon.

### 2.3. Attention and transformer architectures

Vision Transformers [3], hierarchical variants (Swin), and modernised convolutional designs (ConvNeXt [4]) have been transferred to plant pathology, generally reporting marginal gains over strong CNNs at substantially higher data and compute cost. Hybrid designs such as MobileViT [5] are more relevant to the agricultural setting, combining local convolutional inductive bias with global token mixing at mobile parameter budgets. Attention modules (SE, CBAM, coordinate attention)

are frequently bolted onto backbones with reported improvements that are almost never accompanied by an ablation over multiple seeds or a significance test.

### 2.4. Papaya-specific studies

Table 1 summarises the papaya-specific literature along the axes that matter for deployment rather than for leaderboard position. *You must populate this table from your own screening pass*; the column set is the contribution — it is designed so that a reviewer can see at a glance that no prior study occupies the cell that this paper occupies.

### 2.5. Data leakage in applied vision

Near-duplicate leakage is a recognised and repeatedly rediscovered failure mode in applied machine learning, documented in medical imaging (multiple scans per patient split across folds), remote sensing (overlapping tiles) and text (near-duplicate documents)<sup>†</sup>. The standard remedy — grouping at the level of the physically distinct entity and splitting groups, not samples — is well established in those fields and essentially absent from plant pathology. We import it and quantify what it costs.

### 2.6. Calibration, uncertainty and open-set recognition

Modern networks are systematically overconfident [9]. Evidential deep learning [8] replaces the softmax with a Dirichlet distribution over class probabilities, yielding an explicit uncertainty mass in one forward pass — attractive for edge deployment where MC-dropout or deep ensembles are too expensive. Open-set recognition and OOD detection provide the evaluation machinery (AUROC, FPR@95TPR) that lets us state, quantitatively, what the model does when shown something it was never trained on.

### 2.7. Explainability and its evaluation

Grad-CAM [6] and Grad-CAM++ [7] are near-universal in plant-disease papers and near-universally unevaluated. Faithfulness metrics — deletion and insertion AUC, and the pointing game against ground-truth regions — convert explanation from a figure into a measurement. Since expert lesion masks can be annotated on a modest subset (we use  $n = 10$  images), there is no methodological obstacle to doing this properly.

### 2.8. Gap-to-contribution mapping

**Table 2.** Explicit mapping of identified gaps to contributions, experiments and the section reporting each.

| Gap | Closed by                                                          | Contr. | Reported in |
|-----|--------------------------------------------------------------------|--------|-------------|
| G1  | Two-stage leakage audit + group split + published $\rho$           | C1     | §3.5, RQ2   |
| G2  | Three-way background ablation probe                                | C2     | §3.6, RQ3   |
| G3  | Cross-source split + 15-corruption suite                           | C4     | §3.7, RQ4   |
| G4  | Evidential head + open-set split + rejection curve                 | C3,C4  | §4.2, RQ5   |
| G5  | Reliability diagrams, ECE/MCE, Brier, AURC, temperature scaling    | C3     | §5.2, RQ6   |
| G6  | Deletion/insertion AUC, pointing game, stability                   | C5     | §5.4, RQ8   |
| G7  | INT8 latency/RAM/energy on 2 devices; 5 seeds + significance tests | C6     | §5.5, RQ9   |

**Table 1.** Prior papaya (and closely related) leaf disease recognition studies, scored on deployment-relevant methodology. **Y** = reported, **P** = partial/qualitative only, **N** = not reported. Populate from your screening pass; verify every number at the source <sup>†</sup>.

| Study            | Year | Dataset / source (images)            | Classes         | Model                              | Reported acc. | Leak. audit | Group split | Cross-src. | XAI      | Edge lat. | Code     |
|------------------|------|--------------------------------------|-----------------|------------------------------------|---------------|-------------|-------------|------------|----------|-----------|----------|
| [Ref A]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>N</b> | <b>N</b>  | <b>N</b> |
| [Ref B]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>P</b> | <b>N</b>  | <b>N</b> |
| [Ref C]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>N</b> | <b>N</b>  | <b>N</b> |
| [Ref D]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>P</b> | <b>P</b>  | <b>N</b> |
| [Ref E]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>N</b> | <b>N</b>  | <b>Y</b> |
| [Ref F]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>N</b> | <b>N</b>  | <b>N</b> |
| [Ref G]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>P</b> | <b>N</b>  | <b>N</b> |
| [Ref H]          | –    | –                                    | –               | –                                  | –             | <b>N</b>    | <b>N</b>    | <b>N</b>   | <b>N</b> | <b>N</b>  | <b>N</b> |
| <b>This work</b> | 2026 | 3 public sources + curated field set | 6 (+2 open-set) | PapayaFormer (hybrid + evidential) | –             | <b>Y</b>    | <b>Y</b>    | <b>Y</b>   | <b>Y</b> | <b>Y</b>  | <b>Y</b> |

Screening protocol for filling this table: query Scopus/Web of Science/IEEE Xplore/ScienceDirect/Google Scholar with ("papaya" OR "Carica papaya") AND ("leaf disease" OR "disease detection" OR "disease classification") AND ("deep learning" OR CNN OR "machine learning"), 2015–2026; screen title/abstract, then full text; record the exact split procedure verbatim from each paper’s Methods section. Report the count at each stage (Appendix Appendix E).

### 3. MATERIALS: DATA SOURCES, CURATION AND SPLIT DESIGN

#### 3.1. Design requirement: accessibility

A benchmark that cannot be downloaded is not a benchmark. We therefore constrain the study to sources that are (i) publicly downloadable without institutional agreement, (ii) redistributable under a stated licence, and (iii) citable by DOI or a persistent URL. Our own field collection is released under the same constraint.

#### 3.2. Class taxonomy

We adopt a six-class closed set plus a two-class open set (Table 4). The taxonomy is defined by visually separable foliar syndromes confirmable by a plant pathologist from a leaf photograph alone; conditions requiring laboratory confirmation (e.g. specific bacterial species identification) are deliberately merged or excluded, because a label that cannot be reliably assigned from the input modality cannot be learned from it.

**Table 4.** Class taxonomy. Open-set classes appear *only* in the open-set test split and never in training.

| ID                  | Class                      | Defining foliar signature                                             |
|---------------------|----------------------------|-----------------------------------------------------------------------|
| $c_1$               | Healthy                    | Uniform lamina, intact margins, no chlorosis                          |
| $c_2$               | Papaya Ringspot Virus      | Mosaic mottling, vein-clearing, leaf distortion, shoestring lamina    |
| $c_3$               | Anthracnose                | Water-soaked spots enlarging to sunken necrotic lesions, dark margins |
| $c_4$               | Powdery mildew             | Superficial white–grey mycelial coating, adaxial surface              |
| $c_5$               | Bacterial leaf spot        | Small angular translucent lesions, vein-delimited, chlorotic halo     |
| $c_6$               | Mite / mineral deficiency  | Stippling, bronzing, interveinal chlorosis without lesion necrosis    |
| $c_7^{\text{open}}$ | Herbicide / abiotic injury | Marginal scorch, non-progressive, no pathogen structures              |
| $c_8^{\text{open}}$ | Non-papaya leaf            | Any other species (drawn from D4/D5)                                  |

#### 3.3. Field collection protocol (D6)

The curated field subset is what converts this from a re-analysis into a contribution, and it is cheap to collect. Protocol:

- Sites.**  $\geq$  – orchards/homestead plots across  $\geq 3$  districts, sampled across the growing season to capture phenological and illumination variation.
- Devices.**  $\geq 3$  distinct smartphone models spanning price tiers. Device identity is recorded — it becomes a shift variable, not a nuisance.
- Capture.** Natural light only; no reflector, no uniform back-drop; leaf photographed *in situ*. Three captures per leaf at varying angle/distance, all sharing one leaf\_id.
- Metadata** (CSV schema in Appendix Appendix D): image\_id, leaf\_id, plant\_id, plot\_id, district, date, time, device, orientation, illumination\_class, phenostage, label, annotator\_id, confidence.
- Annotation.** Two independent annotators, at least one qualified in plant pathology; disagreements adjudicated by a third. Report Cohen’s  $\kappa$  (two-rater) or Fleiss’  $\kappa$  and the adjudication rate. *Labels without a reported agreement statistic are not evidence.*
- Lesion masks.** On a stratified subset of  $n = -$  images ( $\geq 30$  per class), an expert traces polygon lesion masks. These are used only for XAI evaluation (Section 5.4), never for training.
- Ethics and permissions.** Landowner consent recorded; no persons, faces, house interiors or identifiable property in frame; GPS coarsened to district level before release.



**Table 3.** Candidate public data sources. Every DOI, URL, image count and licence marked <sup>†</sup> **must be verified at the source before submission** — do not copy these fields from any secondary description, including this document.

| ID | Source / repository                                           | Access route               | Images              | Classes present   | Licence      | Role here                |
|----|---------------------------------------------------------------|----------------------------|---------------------|-------------------|--------------|--------------------------|
| D1 | Papaya leaf disease collection (Mendeley Data) <sup>†</sup>   | DOI <sup>†</sup>           | —                   | —                 | <sup>†</sup> | Primary train/test       |
| D2 | Papaya leaf disease collection (Kaggle) <sup>†</sup>          | Kaggle slug <sup>†</sup>   | —                   | —                 | <sup>†</sup> | Primary train/test       |
| D3 | Papaya leaf imagery (Roboflow Universe / GitHub) <sup>†</sup> | URL <sup>†</sup>           | —                   | —                 | <sup>†</sup> | Cross-source test        |
| D4 | PlantVillage [1]                                              | Public mirror <sup>†</sup> | 54,306 <sup>†</sup> | 38 (no papaya)    | <sup>†</sup> | Negative control / OOD   |
| D5 | PlantDoc (field-condition leaves) <sup>†</sup>                | GitHub <sup>†</sup>        | —                   | multi-crop, field | <sup>†</sup> | OOD / background realism |
| D6 | <b>NU-PapayaField</b> (this work)                             | Zenodo DOI (to mint)       | ≥ —                 | 6 + 2 unseen      | CC BY 4.0    | Field test + open-set    |

*Note on D4/D5:* PlantVillage contains no papaya class. This is an asset, not a limitation — it provides a large, label-free OOD probe: a well-behaved papaya classifier must *reject* tomato and potato leaves rather than assign them a papaya disease with high confidence.

### 3.4. Class distribution and imbalance

**Table 5.** Class distribution after curation, per source.  $IR = n_{\max}/n_{\min}$ .

| Class              | D1 | D2 | D3 | D6 | Total | %   |
|--------------------|----|----|----|----|-------|-----|
| Healthy            | —  | —  | —  | —  | —     | —   |
| PRSV               | —  | —  | —  | —  | —     | —   |
| Anthraxnose        | —  | —  | —  | —  | —     | —   |
| Powdery mildew     | —  | —  | —  | —  | —     | —   |
| Bacterial spot     | —  | —  | —  | —  | —     | —   |
| Mite/deficiency    | —  | —  | —  | —  | —     | —   |
| <b>Total</b>       | —  | —  | —  | —  | —     | 100 |
| Imbalance ratio IR |    |    |    |    |       | —   |

### 3.5. The leakage audit

This is the methodological core of PLD-Bench. The objective is to recover, for every image, the identity of the *physical leaf* it depicts, so that splitting can occur at that granularity.

**Stage 1 — exact and perceptual duplicates.** Compute an MD5 digest (exact duplicates) and a 64-bit perceptual hash (pHash, DCT-based) for every image. Two images are Stage-1 linked if their Hamming distance  $d_H \leq \theta_p$  with  $\theta_p = 5$ .

**Stage 2 — semantic near-duplicates.** Embed every image with a frozen self-supervised encoder  $\phi$  (DINOv2 ViT-S/14; CLIP ViT-B/32 as a robustness check). Two images are Stage-2 linked if

$$\cos(\phi(\mathbf{x}_i), \phi(\mathbf{x}_j)) \geq \theta_e, \quad (1)$$

with  $\theta_e$  calibrated on a hand-labelled validation set of — image pairs to achieve precision  $\geq 0.95$  for the relation "same physical leaf". Report the chosen  $\theta_e$ , its precision/recall, and a sensitivity analysis over  $\theta_e \in \{0.90, 0.925, 0.95, 0.975\}$ .

**Stage 3 — grouping.** Treat Stage-1 and Stage-2 links as edges; extract connected components via union-find. Each component is a group  $g$ . Where the source provides an explicit leaf\_id (D6), it overrides the inferred grouping and is used to *measure* the audit's accuracy — a validation that no prior study can perform and that we can, precisely because we collected D6.

### Algorithm 1: Two-stage leaf-instance leakage audit

**Input:** Images  $\{\mathbf{x}_i\}_{i=1}^N$ ; thresholds  $\theta_p, \theta_e$ ; optional native ids

**Output:** Group labels  $\{g_i\}$ ; leakage rate  $\rho$  for any split

```

1 foreach  $i$  do
2    $h_i \leftarrow \text{pHash}(\mathbf{x}_i)$ ;
3    $z_i \leftarrow \phi(\mathbf{x}_i) / \|\phi(\mathbf{x}_i)\|$ 
4 end
5 Init union-find  $U$  over  $\{1..N\}$ 
6 foreach pair  $(i, j)$  with  $d_H(h_i, h_j) \leq \theta_p$  do
7    $U.\text{union}(i, j)$ 
8 end
9 Build ANN index over  $\{z_i\}$  (FAISS, inner product)
10 foreach  $i$  do
11   foreach  $j \in \text{kNN}(z_i, k=20), j \neq i$  do
12     if  $z_i^\top z_j \geq \theta_e$  then
13        $U.\text{union}(i, j)$ 
14     end
15   end
16 end
17 if native ids available then override  $g_i \leftarrow \text{leaf\_id}_i$ 
   and log agreement with  $U$ 
18  $g_i \leftarrow U.\text{find}(i)$ 
19 return  $\{g_i\}$ , and  $\rho$  by Eq. (2) for the split under test

```

**Stage 4 — leakage rate.** For any split  $(\mathcal{T}_{\text{train}}, \mathcal{T}_{\text{test}})$ , define

$$\rho = \frac{1}{|\mathcal{T}_{\text{test}}|} \sum_{i \in \mathcal{T}_{\text{test}}} \mathbb{I}[\exists j \in \mathcal{T}_{\text{train}} : g_i = g_j]. \quad (2)$$

We report  $\rho$  for the random split (expected to be large) and for the group split (exactly 0 by construction). We propose  $\rho$  as a mandatory reporting statistic for any plant-disease dataset paper.

### 3.6. The Clever-Hans background probe

Segment the leaf in every test image (SAM or a fine-tuned U-Net; report segmentation IoU on a manually masked subset), then construct three variants:  $\mathcal{V}_{\text{full}}$  (unmodified),  $\mathcal{V}_{\text{leaf}}$

(background set to a neutral constant),  $\mathcal{V}_{\text{bg}}$  (leaf region set to a neutral constant). Evaluate the *same trained model* on all three and report

$$\Delta_{\text{bg}} = \text{Acc}(\mathcal{V}_{\text{bg}}) - \frac{1}{K}, \quad \Delta_{\text{leaf}} = \text{Acc}(\mathcal{V}_{\text{full}}) - \text{Acc}(\mathcal{V}_{\text{leaf}}). \quad (3)$$

$\Delta_{\text{bg}} \gg 0$  is direct evidence that the dataset — not the model — is broken: the background alone carries class information. A large  $\Delta_{\text{leaf}}$  indicates dependence on context that will not transfer. Both quantities should be reported for every dataset used, ours included.

### 3.7. Split design

**Table 6.** Split design. S1 exists solely to quantify the inflation that S2 removes.

| ID | Name             | Construction                                                                                              | $\rho$ |
|----|------------------|-----------------------------------------------------------------------------------------------------------|--------|
| S1 | Random (legacy)  | Uniform 70/15/15 over images, ignoring groups. Reproduces prior practice.                                 | —      |
| S2 | Group-stratified | Group-stratified 5-fold over $g$ ; class ratios preserved; no group spans folds. <b>Primary protocol.</b> | 0      |
| S3 | Cross-source     | Train on D1UD2, test on D3UD6. Measures acquisition shift.                                                | 0      |
| S4 | Corruption       | S2 test set $\times$ 15 corruptions $\times$ 5 severities.                                                | 0      |
| S5 | Open-set         | S2 train ( $c_1$ – $c_6$ ); test adds $c_7, c_8$ as reject targets.                                       | 0      |
| S6 | Low-data         | S2 train subsampled to $\{1, 5, 10, 25, 50\}\%$ by group.                                                 | 0      |

### 3.8. Preprocessing and augmentation

Images are resized with aspect-preserving letterboxing to  $224 \times 224$  (and  $384 \times 384$  for the high-resolution ablation) and normalised with ImageNet statistics. Augmentation choices are justified individually in Table 7; note in particular the *excluded* operations, which is the part reviewers examine.

**Table 7.** Augmentation policy. Exclusions are argued from disease phenomenology, not convention.

| Operation                          | Use      | Rationale                                                                                                |
|------------------------------------|----------|----------------------------------------------------------------------------------------------------------|
| Horizontal/vertical flip           | <b>Y</b> | Leaf orientation is arbitrary.                                                                           |
| Rotation $\pm 30^\circ$            | <b>Y</b> | Handheld capture angle varies.                                                                           |
| Random resized crop (0.7–1.0)      | <b>Y</b> | Framing distance varies; preserves lesion scale range.                                                   |
| Brightness/contrast $\pm 20\%$     | <b>Y</b> | Models illumination variation.                                                                           |
| Mild blur, JPEG $q \in [60, 95]$   | <b>Y</b> | Matches phone capture artefacts.                                                                         |
| Cutout ( $\leq 1$ hole, 16%)       | <b>Y</b> | Discourages single-lesion reliance.                                                                      |
| MixUp / CutMix                     | <b>P</b> | Ablated separately: label mixing can be harmful when lesion area is small.                               |
| Strong hue jitter ( $> 15^\circ$ ) | <b>N</b> | <b>Destroys the chlorosis/necrosis colour cue that separates <math>c_2</math> from <math>c_6</math>.</b> |
| Grayscale conversion               | <b>N</b> | Same reason; colour is diagnostic here.                                                                  |
| Aggressive elastic warp            | <b>N</b> | Corrupts vein geometry used for vein-clearing detection.                                                 |
| Vertical shear $> 10^\circ$        | <b>N</b> | Distorts lesion morphology (angular vs. circular) that distinguishes $c_5$ from $c_3$ .                  |

All augmentation is applied *after* splitting and only to the training fold. Test-time augmentation, if used, is reported separately and never conflated with the base result.

## 4. METHOD

### 4.1. Problem formulation

Let  $\mathcal{D} = \{(\mathbf{x}_i, y_i, g_i)\}_{i=1}^N$  with  $\mathbf{x}_i \in \mathbb{R}^{3 \times H \times W}$ ,  $y_i \in \{1, \dots, K\}$ ,  $K = 6$ , and  $g_i$  the leaf-instance group. We seek  $f_\theta$  minimising expected risk on a *shifted* test distribution  $q \neq p$ , subject to two operational constraints absent from the standard formulation:

$$(i) \text{ Calibration: } \Pr(y = \hat{y} \mid \hat{p} = c) = c \quad \forall c \in [0, 1], \quad (4)$$

$$(ii) \text{ Rejection: } f_\theta \text{ abstains when } \mathbf{x} \sim q_{\text{open}} \notin \{c_1 \dots c_K\}. \quad (5)$$

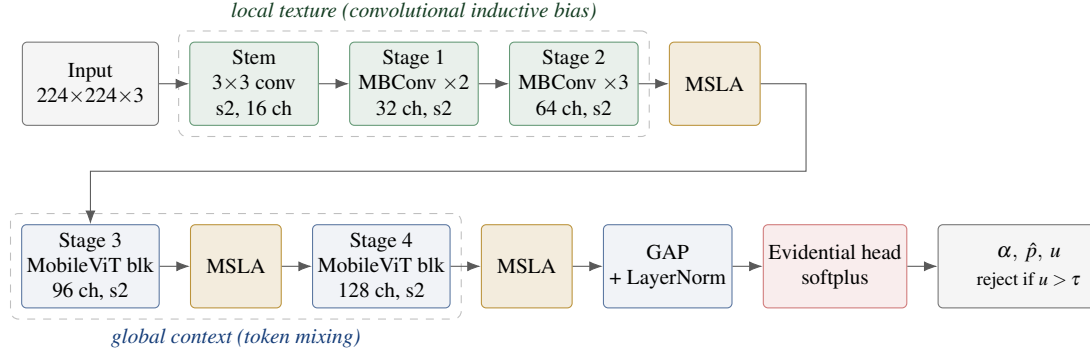
A model that satisfies (i) and (ii) at 92% accuracy is more deployable than one at 99% that satisfies neither, and the evaluation in Section 6 is designed to make that trade-off visible.

### 4.2. Architecture overview

PapayaFormer is a four-stage hierarchical encoder. Stages 1–2 are inverted-residual convolutional blocks (local texture: mycelium granularity, lesion margins). Stages 3–4 are MobileViT-style blocks that unfold feature maps into patches, apply lightweight self-attention for global context (mosaic motting is a *global* pattern that local receptive fields miss), and fold back. An MSLA module is inserted at the end of stages 2, 3 and 4. The head is evidential rather than softmax.

### 4.3. Multi-Scale Lesion Attention (MSLA)

Papaya foliar symptoms occupy radically different spatial scales within one image: bacterial spots are a few pixels across, PRSV mosaic spans the entire lamina. A fixed receptive field is therefore mis-specified by construction. Given  $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$ , MSLA computes three dilated branches, fuses them into a



Multi-scale supervision: the MSLA masks at stages 2, 3 and 4 are regularised jointly (Eq. 15), so the network is penalised for placing attention outside the leaf — even when the background is class-correlated and attending to it would reduce training loss.

**Figure 1.** PapayaFormer. Convolutional stages capture punctate and textural symptoms (powdery mildew granularity, necrotic lesion margins); token-mixing stages capture spatially extended symptoms (PRSV mosaic, interveinal chlorosis) that exceed a local receptive field. MSLA modules re-weight features by multi-dilation spatial evidence and channel importance. The evidential head yields a Dirichlet posterior, giving class probabilities *and* an explicit uncertainty mass  $u$  in a single forward pass — a requirement for rejection on an edge device, where ensembling or MC-dropout is too expensive.

spatial evidence map, and modulates channels:

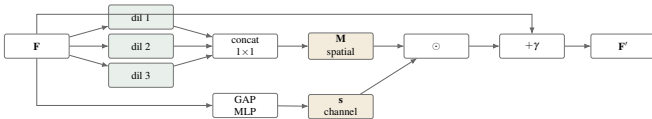
$$\mathbf{F}_d = \text{BN}(\text{Conv}_{3 \times 3}^{\text{dil}=d}(\mathbf{F})), \quad d \in \{1, 2, 3\}, \quad (6)$$

$$\mathbf{M} = \sigma\left(\text{Conv}_{1 \times 1}[\mathbf{F}_1 \parallel \mathbf{F}_2 \parallel \mathbf{F}_3]\right) \in \mathbb{R}^{1 \times H \times W}, \quad (7)$$

$$\mathbf{s} = \sigma\left(W_2 \delta(W_1 \text{GAP}(\mathbf{F}))\right) \in \mathbb{R}^C, \quad (8)$$

$$\mathbf{F}' = \mathbf{F} + \gamma(\mathbf{F} \odot \mathbf{M} \odot \mathbf{s}), \quad (9)$$

where  $\sigma$  is the logistic function,  $\delta$  is GELU,  $W_1 \in \mathbb{R}^{C/r \times C}$ ,  $W_2 \in \mathbb{R}^{C \times C/r}$  with reduction  $r = 8$ , and  $\gamma$  is a learnable scalar initialised to 0 so that MSLA starts as identity (stabilising fine-tuning from ImageNet weights). Parameter overhead is  $O(C^2/r)$  per module; the exact count is reported in Table 16.



**Figure 2.** MSLA internals. The residual path with learnable  $\gamma$  initialised at zero makes the module a no-op at initialisation, which is what allows it to be inserted into a pretrained backbone without destabilising early fine-tuning.

#### 4.4. Evidential head, calibrated confidence and rejection

Replace the softmax with a non-negative evidence vector  $\mathbf{e} = \text{softplus}(f_\theta(\mathbf{x})) \in \mathbb{R}_+^K$ , inducing a Dirichlet posterior over the simplex:

$$\alpha = \mathbf{e} + \mathbf{1}, \quad S = \sum_{k=1}^K \alpha_k, \quad \hat{p}_k = \frac{\alpha_k}{S}, \quad u = \frac{K}{S}. \quad (10)$$

$u \in (0, 1]$  is total evidential uncertainty: it is high both for ambiguous in-distribution inputs and for inputs unlike anything seen in training, which is exactly the abstention signal a field application needs. Prediction with rejection:

$$\hat{y}(\mathbf{x}) = \begin{cases} \arg \max_k \hat{p}_k, & u(\mathbf{x}) \leq \tau, \\ \text{reject / refer to expert}, & u(\mathbf{x}) > \tau, \end{cases} \quad (11)$$

with  $\tau$  fixed on the *validation* fold at 95% TPR for in-distribution acceptance and then frozen. Reporting a threshold tuned on the test set is a disqualifying error and we state explicitly that we do not do it.

#### 4.5. Training objective

The composite loss addresses imbalance, evidential fitting, augmentation consistency and attention regularisation:

$$\mathcal{L}_{\text{EDL}} = \sum_{k=1}^K y_k (\psi(S) - \psi(\alpha_k)) + \lambda_t \text{KL}(\text{Dir}(\tilde{\alpha}) \parallel \text{Dir}(\mathbf{1})), \quad (12)$$

$$\mathcal{L}_{\text{CB}} = -\frac{1-\beta}{1-\beta^{n_k}} (1-\hat{p}_k)^{\gamma_f} \log \hat{p}_k, \quad (13)$$

$$\mathcal{L}_{\text{con}} = \|\hat{p}(\mathcal{A}_1(\mathbf{x})) - \hat{p}(\mathcal{A}_2(\mathbf{x}))\|_2^2, \quad (14)$$

$$\mathcal{L}_{\text{attn}} = \frac{1}{HW} \|\mathbf{M}\|_1 + \mu \text{TV}(\mathbf{M}), \quad (15)$$

$$\mathcal{L} = \mathcal{L}_{\text{EDL}} + \lambda_1 \mathcal{L}_{\text{CB}} + \lambda_2 \mathcal{L}_{\text{con}} + \lambda_3 \mathcal{L}_{\text{attn}} \quad (16)$$

where  $\psi$  is the digamma function,  $\tilde{\alpha} = \mathbf{y} + (1 - \mathbf{y}) \odot \alpha$  removes the evidence of the true class before penalising,  $\lambda_t = \min(1, t/T_{\text{anneal}})$  anneals the KL term over the first  $T_{\text{anneal}}$  epochs (without annealing, EDL collapses to maximal uncertainty early in training — a failure we document in Appendix B),  $n_k$  is the class frequency,  $\beta = 0.9999$ ,  $\gamma_f = 2$ ,  $\mathcal{A}_1, \mathcal{A}_2$  are independent augmentations, and TV is total variation encouraging spatially coherent rather than speckled attention. Sensitivity to  $\lambda_1, \lambda_2, \lambda_3$  is reported as a grid in Appendix B; unreported loss weights are a standard reviewer objection and we pre-empt it.

#### 4.6. Baselines

All baselines use identical splits, augmentation, schedule, seeds and early-stopping criteria; only the architecture differs. Anything else is not a comparison.

- **Classical:** GLCM+LBP+colour-moment features  $\rightarrow$  RBF-SVM; and  $\rightarrow$  Random Forest.

**Algorithm 2:** CLEAR-Papaya training and evaluation

---

**Input:**  $\mathcal{D}$ , groups  $\{g_i\}$ , seeds  $\mathcal{S} = \{0..4\}$   
**Output:** Per-fold metrics, CIs, significance tests

```

1  $\{g_i\}, \rho \leftarrow \text{LEAKAGEAUDIT}(\mathcal{D})$  // Alg. 1
2 foreach seed  $s \in \mathcal{S}$  do
3   folds  $\leftarrow \text{GroupStratifiedKFold}(\mathcal{D}, g, k=5, \text{seed } s)$ 
4   foreach fold  $(\mathcal{T}_{tr}, \mathcal{T}_{va}, \mathcal{T}_{te})$  do
5     init  $\theta$  from ImageNet weights;  $\gamma \leftarrow 0$ 
6     for epoch = 1.. $E$  do
7       warmup (5 ep., linear) then cosine decay
8       optimise the composite objective with AdamW, EMA 0.999
9       early-stop on validation macro-F1, patience 15
10    end
11     $\tau \leftarrow$  quantile of  $u$  on  $\mathcal{T}_{va}$  at TPR 0.95
12    evaluate on  $\mathcal{T}_{te}$ , S3–S6, corruption suite, XAI suite
13  end
14 end
15 aggregate: mean  $\pm$  sd, BCa bootstrap 95% CI ( $B=10^4$ )
16 test:  $5 \times 2$ cv paired  $t$ , Wilcoxon, McNemar; Holm correction

```

---

- **CNN:** ResNet-50, DenseNet-121, EfficientNet-B0, MobileNetV3-Large, ConvNeXt-Tiny [4].
- **Transformer:** ViT-B/16 [3], Swin-T, DeiT-S, MobileViT-S [5].
- **Attention-augmented CNN:** ResNet-50+SE, ResNet-50+CBAM.
- **Ours:** PapayaFormer (+ variants in Table 16).

**4.7. Hyperparameters****Table 8.** Training configuration. Identical across all architectures unless stated; the search protocol is in Appendix B.

| Setting                       | Value                                                   |
|-------------------------------|---------------------------------------------------------|
| Input resolution              | $224 \times 224$ (ablation: 384)                        |
| Optimiser                     | AdamW, $\beta = (0.9, 0.999)$ , wd 0.05                 |
| LR schedule                   | 5-epoch linear warmup $\rightarrow$ cosine              |
| Peak LR                       | $3 \times 10^{-4}$ (head $10 \times$ )                  |
| Batch size                    | 32 (grad. accum. to 128 effective)                      |
| Epochs / early stop           | 100 / patience 15 on val macro-F1                       |
| Label smoothing               | 0.05 (softmax baselines only)                           |
| EMA decay                     | 0.999                                                   |
| Loss weights                  | $\lambda_1=1.0, \lambda_2=0.5, \lambda_3=0.01, \mu=0.1$ |
| KL anneal $T_{\text{anneal}}$ | 20 epochs                                               |
| Precision                     | AMP fp16                                                |
| Seeds                         | $\{0, 1, 2, 3, 4\}$ , all results $\pm$ sd              |
| Framework                     | PyTorch $-$ , timm $-$ , CUDA $-$                       |

**5. EXPERIMENTAL SETUP****5.1. Environment****Table 9.** Compute environment. Report exactly; "a GPU" is not a specification.

| Component               | Specification                                         |
|-------------------------|-------------------------------------------------------|
| Training accelerator    | $-$ (e.g. NVIDIA T4 16 GB, Colab)                     |
| CPU / RAM               | $-$                                                   |
| OS / driver / CUDA      | $-$                                                   |
| Python / PyTorch / timm | $-$                                                   |
| Edge device A           | $-$ (mid-range Android, SoC $-$ )                     |
| Edge device B           | Raspberry Pi 4B 4 GB, ARM Cortex-A72                  |
| Total training compute  | $-$ GPU-hours; est. $-$ kWh, $-$ kg CO <sub>2</sub> e |

**5.2. Metrics**

Accuracy alone is uninformative under imbalance and useless under rejection. We report:

**Discrimination.** Overall accuracy; macro precision, recall and  $F_1$ ; per-class  $F_1$ ; balanced accuracy; Matthews correlation coefficient

$$\text{MCC} = \frac{c \cdot s - \sum_k p_k t_k}{\sqrt{(s^2 - \sum_k p_k^2)(s^2 - \sum_k t_k^2)}}, \quad (17)$$

with  $c$  correct predictions,  $s$  total,  $p_k$  predicted and  $t_k$  true counts; Cohen’s  $\kappa$ ; one-vs-rest AUROC and AUPRC (macro and weighted).

**Calibration.** Expected and maximum calibration error over  $M = 15$  equal-mass bins,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (18)$$

$\text{MCE} = \max_m |\text{acc}(B_m) - \text{conf}(B_m)|$ , plus the multiclass Brier score  $\text{BS} = \frac{1}{n} \sum_i \sum_k (\hat{p}_{ik} - y_{ik})^2$ , reliability diagrams, and post-hoc temperature scaling as a comparator.

**Selective prediction.** Risk-coverage curves and AURC; accuracy at coverage  $\{100, 90, 80, 70\}\%$ . This is the number a deployment decision actually turns on: *what accuracy do we get on the images we choose to answer?*

**Open-set / OOD.** AUROC, AUPR-Out and FPR@95TPR using  $u$  as the score, against  $c_7$ ,  $c_8$  and D4/D5.

**Robustness.** Mean corruption accuracy across 15 corruptions  $\times$  5 severities, and relative degradation  $\text{rCE} = 1 - \text{Acc}_{\text{corr}} / \text{Acc}_{\text{clean}}$  [10].

**Efficiency.** Parameters (M), GFLOPs at  $224^2$ , peak training memory, and edge metrics in Section 5.5.

**5.3. Statistical protocol**

Every comparison in Section 6 is accompanied by an inferential statement. Specifically:

- All results are mean  $\pm$  sd over 5 seeds  $\times$  5 folds.
- 95% confidence intervals by BCa bootstrap,  $B = 10,000$  resamples *at group level*, not image level — bootstrapping images inside a leaked group manufactures false precision.
- Pairwise model comparison:  $5 \times 2$ cv paired  $t$ -test [11] for accuracy; McNemar’s test on the shared test predictions for paired error patterns.



- Multi-model comparison: Friedman test followed by Nemenyi post-hoc with a critical-difference diagram [12].
- Multiplicity: Holm–Bonferroni across the family of comparisons within each research question; the family is stated explicitly in each table caption.
- Effect size: Cliff’s  $\delta$  (non-parametric) alongside every  $p$ -value. A significant  $p$  with  $\delta < 0.147$  is reported as negligible and described as such in the text.
- Pre-registration: the hypothesis set (RQ1–RQ10), the primary metric (macro- $F_1$  on S2) and the analysis plan are fixed before results are unblinded; deviations are logged in Appendix A.

#### 5.4. Explainability evaluation protocol

Four attribution methods — Grad-CAM [6], Grad-CAM++ [7], Score-CAM and SHAP (partition explainer) — are compared on three axes:

1. **Faithfulness.** Deletion AUC (progressively mask the top- $k\%$  most salient pixels; *lower is better*) and insertion AUC (progressively reveal them onto a blurred baseline; *higher is better*), at 1% steps.
2. **Localisation.** Pointing game against the expert lesion masks of Section 3.3: hit if the attribution argmax lies inside a lesion polygon. Also report attribution mass inside the leaf mask,  $\Omega = \sum_{\text{leaf}} A / \sum A$  — a direct, quantitative version of the Clever-Hans test.
3. **Stability.** Mean rank correlation of attribution maps under small input perturbations (Gaussian  $\sigma = 0.01$ , 10 draws). An explanation that changes under imperceptible noise is not an explanation.

Sanity checks (model-parameter randomisation and label randomisation) are run and reported; an attribution method that survives randomisation is excluded from the analysis.

#### 5.5. Edge deployment protocol

Export to ONNX and TensorFlow Lite; quantise to FP16 and INT8 (post-training, 500-image calibration set drawn from training data only). On each device, measure over 200 runs after 20 warm-up runs, single-threaded and 4-threaded: median and p95 latency, peak RSS, model file size, and energy per inference (device A: battery-delta method over 1000 inferences; device B: USB inline power meter). Report the paired accuracy of every quantised variant on the identical S2 test set, per class — aggregate accuracy hides the fact that quantisation typically destroys the minority class first.

## 6. RESULTS

### Status

Every table in this section is a pre-registered scaffold. Cell contents (–) are to be filled from your runs. Do not alter the column structure after unblinding — the structure is the pre-registration, and keeping it fixed is what makes the eventual numbers credible.

#### 6.1. RQ1 — Baseline comparison under the leakage-free protocol

#### 6.2. RQ2 — The cost of leaky splitting

This is the result the paper is built around. Identical models, identical data, identical hyperparameters; only the split

procedure changes.

**Table 11.** Accuracy inflation from random splitting.  $\Delta = S1 - S2$ . Family: 4 paired comparisons, Holm-corrected.

| Model           | S1 random<br>( $\rho=-$ ) | S2 group<br>( $\rho=0$ ) | $\Delta$ (pp) | $p$ |
|-----------------|---------------------------|--------------------------|---------------|-----|
| ResNet-50       | –                         | –                        | –             | –   |
| EfficientNet-B0 | –                         | –                        | –             | –   |
| MobileViT-S     | –                         | –                        | –             | –   |
| PapayaFormer    | –                         | –                        | –             | –   |

**[FIGURE SLOT 2]** Grouped bar chart: accuracy under S1 vs. S2 for every model, with 95% CI whiskers; secondary axis showing  $\rho$  per split.

**Figure 3.** Reported accuracy as a function of split procedure. Note that the *ranking* of models may also change, which would mean prior architectural conclusions in this literature do not survive the correction.

#### 6.3. RQ3 — Clever-Hans background probe

**Table 12.** Background ablation.  $\text{Acc}(\%_{\text{bg}})$  substantially above chance ( $1/K = 16.7\%$ ) indicates dataset-level background confounding.

| Dataset    | Full (%) | Leaf-only<br>(%) | Bg-only<br>(%) | $\Delta_{\text{bg}}$ (pp) |
|------------|----------|------------------|----------------|---------------------------|
| D1         | –        | –                | –              | –                         |
| D2         | –        | –                | –              | –                         |
| D3         | –        | –                | –              | –                         |
| D6 (field) | –        | –                | –              | –                         |

#### 6.4. RQ4 — Distribution shift and corruption

**Table 13.** Cross-source transfer (S3) and corruption robustness (S4). rCE lower is better.

| Model           | In-dist. $F_1$ | Cross-src.<br>$F_1$ | Drop (pp) | rCE |
|-----------------|----------------|---------------------|-----------|-----|
| ResNet-50       | –              | –                   | –         | –   |
| EfficientNet-B0 | –              | –                   | –         | –   |
| Swin-T          | –              | –                   | –         | –   |
| MobileViT-S     | –              | –                   | –         | –   |
| PapayaFormer    | –              | –                   | –         | –   |

**Table 10.** Primary results on split S2 (group-stratified,  $p = 0$ ). Mean  $\pm$  sd over 5 seeds  $\times$  5 folds; 95% BCa CI for macro- $F_1$  in brackets. Comparison family for Holm correction: all rows vs. PapayaFormer ( $m = -$  comparisons).  $\dagger$  significant at  $\alpha = 0.05$  after correction;  $\delta$  = Cliff’s delta vs. PapayaFormer.

| Model                      | Acc. (%) | Macro- $F_1$ | MCC | $\kappa$ | AUROC | ECE $\downarrow$ | Params (M) | GFLOPs | $p$ | $\delta$ |
|----------------------------|----------|--------------|-----|----------|-------|------------------|------------|--------|-----|----------|
| GLCM+LBP $\rightarrow$ SVM | -        | -            | -   | -        | -     | -                | -          | -      | -   | -        |
| GLCM+LBP $\rightarrow$ RF  | -        | -            | -   | -        | -     | -                | -          | -      | -   | -        |
| ResNet-50                  | -        | -            | -   | -        | -     | -                | 25.6       | 4.1    | -   | -        |
| DenseNet-121               | -        | -            | -   | -        | -     | -                | 8.0        | 2.9    | -   | -        |
| EfficientNet-B0            | -        | -            | -   | -        | -     | -                | 5.3        | 0.4    | -   | -        |
| MobileNetV3-L              | -        | -            | -   | -        | -     | -                | 5.4        | 0.2    | -   | -        |
| ConvNeXt-Tiny              | -        | -            | -   | -        | -     | -                | 28.6       | 4.5    | -   | -        |
| ResNet-50 + SE             | -        | -            | -   | -        | -     | -                | 28.1       | 4.1    | -   | -        |
| ResNet-50 + CBAM           | -        | -            | -   | -        | -     | -                | 28.1       | 4.1    | -   | -        |
| ViT-B/16                   | -        | -            | -   | -        | -     | -                | 86.6       | 17.6   | -   | -        |
| DeiT-S                     | -        | -            | -   | -        | -     | -                | 22.1       | 4.6    | -   | -        |
| Swin-T                     | -        | -            | -   | -        | -     | -                | 28.3       | 4.5    | -   | -        |
| MobileViT-S                | -        | -            | -   | -        | -     | -                | 5.6        | 2.0    | -   | -        |
| <b>PapayaFormer (ours)</b> | -        | - [-, -]     | -   | -        | -     | -                | -          | -      | -   | -        |

**[FIGURE SLOT 3]** Heatmap: accuracy across 15 corruption types (rows)  $\times$  5 severities (columns) for PapayaFormer, with a difference panel vs. the strongest baseline.**Figure 4.** Corruption profile. The clinically relevant corruptions here are motion blur, low light and JPEG compression — the failure modes of a handheld phone in a field at dusk.**[FIGURE SLOT 4]** (a) Reliability diagrams (ours vs. best baseline, with gap bars); (b) risk–coverage curves with AURC annotated.**Figure 5.** Calibration and selective risk. A deployed triage tool is used at partial coverage; panel (b) is therefore the operationally meaningful plot.

## 6.5. RQ5 — Open-set rejection

**Table 14.** Open-set detection using uncertainty  $u$  (ours) or max-softmax (baselines).  $\tau$  fixed on validation.

| Method            | AUROC $\uparrow$ | AUPR-Out $\uparrow$ | FPR@95 $\downarrow$ | Closed-set $F_1$ |
|-------------------|------------------|---------------------|---------------------|------------------|
| Max-softmax       | -                | -                   | -                   | -                |
| Energy score      | -                | -                   | -                   | -                |
| MC-dropout (20)   | -                | -                   | -                   | -                |
| Deep ensemble (5) | -                | -                   | -                   | -                |
| Evidential (ours) | -                | -                   | -                   | -                |

## 6.6. RQ6 — Calibration and selective prediction

**Table 15.** Calibration before and after temperature scaling, and selective-prediction accuracy at fixed coverage.

| Model           | ECE | MCE | Brier | AURC $\downarrow$ | Acc@80% cov. |
|-----------------|-----|-----|-------|-------------------|--------------|
| ResNet-50       | -   | -   | -     | -                 | -            |
| + temp. scaling | -   | -   | -     | -                 | -            |
| MobileViT-S     | -   | -   | -     | -                 | -            |
| PapayaFormer    | -   | -   | -     | -                 | -            |

## 6.7. RQ7 — Ablation study

**Table 16.** Component ablation on S2. Each row removes exactly one component from the full model.  $\Delta F_1$  vs. full;  $p$  from  $5 \times 2$ cv paired  $t$ -test, Holm-corrected within this family.

| Variant                                    | Macro- $F_1$ | $\Delta$ | Params (M) | $p$ |
|--------------------------------------------|--------------|----------|------------|-----|
| Full PapayaFormer                          | -            | -        | -          | -   |
| – MSLA (all stages)                        | -            | -        | -          | -   |
| – MSLA (stage 4 only kept)                 | -            | -        | -          | -   |
| – evidential head ( $\rightarrow$ softmax) | -            | -        | -          | -   |
| – class-balanced focal                     | -            | -        | -          | -   |
| – consistency loss                         | -            | -        | -          | -   |
| – attention regulariser                    | -            | -        | -          | -   |
| – transformer stages ( $\rightarrow$ conv) | -            | -        | -          | -   |
| – ImageNet init (scratch)                  | -            | -        | -          | -   |
| + MixUp/CutMix                             | -            | -        | -          | -   |
| + input $384^2$                            | -            | -        | -          | -   |

## 6.8. RQ8 — Explanation faithfulness

**Table 17.** Quantitative XAI evaluation on the  $n = -$  expert-masked subset.  $\Omega$  = attribution mass inside the leaf.

| Method                     | Del. AUC<br>↓ | Ins. AUC ↑ | Pointing<br>(%) | $\Omega$ |
|----------------------------|---------------|------------|-----------------|----------|
| Grad-CAM                   | —             | —          | —               | —        |
| Grad-CAM++                 | —             | —          | —               | —        |
| Score-CAM                  | —             | —          | —               | —        |
| SHAP (parti-<br>tion)      | —             | —          | —               | —        |
| MSLA mask (in-<br>trinsic) | —             | —          | —               | —        |

[FIGURE SLOT 5] Qualitative grid: 2 examples per class  $\times$  {input, expert mask, Grad-CAM++, MSLA mask}. **Include at least two failure cases** — a paper whose qualitative figure contains only successes is read as selective reporting.

**Figure 6.** Attributions with expert lesion masks for reference. Failure cases are shown deliberately and analysed in Section 7.

## 6.9. RQ9 — Edge deployment

## 6.10. RQ10 — Low-data regime

[FIGURE SLOT 6] Macro- $F_1$  vs. training-set fraction {1, 5, 10, 25, 50, 100}% (log  $x$ ), shaded 95% CI, one curve per model.

**Figure 7.** Data efficiency. This curve is what a research group in a new papaya-growing region needs in order to estimate how much local data they must collect.

[FIGURE SLOT 7] (a) Normalised confusion matrix on S2; (b) t-SNE/UMAP of penultimate features coloured by class, with open-set samples overlaid in grey.

**Figure 8.** Error structure. Report which confusions are pathologically meaningful (e.g.  $c_5 \leftrightarrow c_3$  early-stage lesions are genuinely ambiguous to human experts too) and which are not.

# 7. DISCUSSION

## 7.1. Interpreting the leakage result

(Write after RQ2 is filled.) Frame the finding in terms of what it does and does not imply. If  $\Delta$  in Table 11 is large, the correct claim is: **published papaya leaf disease accuracies are not comparable to ours and are not estimates of field**

**performance.** The incorrect claim — which reviewers will look for and punish — is that prior authors acted in bad faith. They followed a convention. The contribution is retiring the convention, and the tone of this subsection should reflect that.

State explicitly which prior conclusions survive: if model *ranking* is preserved under S2, then architectural findings in the literature remain informative even though the absolute numbers do not. If ranking changes, say so plainly and show the rank-correlation coefficient.

## 7.2. What the background probe implies for dataset construction

(Write after RQ3.) If  $\Delta_{bg}$  is large on the public sources but small on D6, the practical recommendation follows directly: benchmark datasets for plant pathology must be collected with class-independent backgrounds, or must publish  $\Delta_{bg}$  so that users can discount accordingly. Propose  $\rho$  and  $\Delta_{bg}$  as a two-number dataset-quality label, analogous to a datasheet entry.

## 7.3. Deployment implications

Combine RQ5, RQ6 and RQ9 into a single operational statement of the form: *at coverage X%, the system achieves Y% accuracy with ECE Z, rejecting W% of inputs to human review, at Lms per inference on a \$120 phone.* That sentence is the paper’s practical payload. Then discuss the referral workflow: what happens to rejected images, who reviews them, and what the expected expert workload is per 1000 farmer queries.

## 7.4. Agronomic relevance

Discuss which confusions are agronomically costly. Confusing  $c_2$  (PRSV, requiring roguing of the entire plant) with  $c_6$  (mite/deficiency, requiring a nutrient or miticide intervention) has an asymmetric cost: a false PRSV call destroys a healthy plant, a missed PRSV call seeds an epidemic. Report a cost-sensitive evaluation with an explicit cost matrix elicited from an agronomist, not a symmetric accuracy. This subsection is where a plant-science reviewer decides whether the paper was written by people who understand the crop.

## 7.5. Comparison with the state of the art

**Table 19.** Positioning against prior work. Note the deliberate columns: absolute accuracy is *not* the comparison axis, because prior numbers are computed under a different (leaky) protocol and are not commensurable.

| Study                             | Protocol under which accuracy was measured | Acc. (%) | Comparable? |
|-----------------------------------|--------------------------------------------|----------|-------------|
| [Ref A]                           | Random split, single source, no audit      | —        | <b>N</b>    |
| [Ref B]                           | Random split, single source, no audit      | —        | <b>N</b>    |
| [Ref C]                           | —                                          | —        | —           |
| Ours (S1, for comparability only) | Random split, reproducing legacy protocol  | —        | <b>Y</b>    |
| Ours (S2, primary)                | Group split, $\rho = 0$                    | —        | —           |

Reporting our own S1 number is what makes this table honest: it lets a reader compare like with like, and then see what the correction costs.

**Table 18.** Measured edge performance. Latency is median (p95) over 200 runs after warm-up. Accuracy is on the identical S2 test set; per-class deltas in Appendix [Appendix C](#).

| Model / precision    | Size (MB) | Android lat. ms<br>(1thr) | Android lat. ms<br>(4thr) | RPi4 lat. ms | Peak RAM<br>(MB) | Energy<br>(mJ/inf.) | Acc. (%) |
|----------------------|-----------|---------------------------|---------------------------|--------------|------------------|---------------------|----------|
| PapayaFormer FP32    | —         | —                         | —                         | —            | —                | —                   | —        |
| PapayaFormer FP16    | —         | —                         | —                         | —            | —                | —                   | —        |
| PapayaFormer INT8    | —         | —                         | —                         | —            | —                | —                   | —        |
| MobileNetV3-L INT8   | —         | —                         | —                         | —            | —                | —                   | —        |
| EfficientNet-B0 INT8 | —         | —                         | —                         | —            | —                | —                   | —        |
| ResNet-50 INT8       | —         | —                         | —                         | —            | —                | —                   | —        |

## 8. THREATS TO VALIDITY AND LIMITATIONS

### 8.1. Internal validity

Group inference by embedding similarity is imperfect; a mis-grouped pair either splits one leaf into two groups (harmless, conservative) or merges two leaves (slightly reduces effective sample diversity). We bound the error using D6’s ground-truth `leaf_id` and report precision/recall of the inferred grouping. Hyperparameters were tuned on validation folds only; the test folds were touched once, at the end.

### 8.2. External validity

Results are conditioned on the cultivars, districts, seasons and devices sampled. Papaya symptom expression varies with cultivar and with co-infection; a model trained on Bangladeshi field data should not be assumed transferable to, e.g., Brazilian commercial orchards without re-evaluation. We report the low-data curve (RQ10) precisely so that others can estimate the re-collection cost.

### 8.3. Construct validity

Photographic diagnosis is a proxy for laboratory confirmation. Labels are expert visual assessments with reported inter-annotator agreement, not PCR- or ELISA-confirmed diagnoses. For PRSV in particular, serological confirmation on a subset would strengthen the label construct; we report the confirmed fraction (—) and treat the remainder as expert-consensus labels.

### 8.4. Conclusion validity

Five seeds  $\times$  five folds is adequate for the effect sizes we expect but underpowered for detecting differences below  $\approx$  — pp; we report the minimum detectable effect from a power analysis rather than interpreting null results as equivalence.

### 8.5. Limitations

1. Class  $c_5$  (bacterial leaf spot) is defined at syndrome rather than species level; species-level identification is not achievable from RGB imagery.
2. Multi-label reality: a leaf can carry two pathogens simultaneously. This work uses single-label classification; we quantify the fraction of images the annotators flagged as co-infected (—%) and exclude them, and we identify multi-label extension as necessary future work rather than a minor detail.
3. Severity is not modelled — only presence. Severity grading is what determines the intervention.

4. The field set is geographically concentrated; nationwide and cross-country collection remains open.

5. Energy measurement on device A uses a battery-delta method with  $\pm$ —% uncertainty.

## 9. CONCLUSION AND FUTURE WORK

(Write last; mirror the abstract, no new claims.) Template: We showed that papaya leaf disease recognition accuracy reported in the literature is measured under a protocol that permits near-duplicate leakage and background confounding, and we quantified both. We released PLD-Bench, a leakage-audited protocol with published  $\rho$  and  $\Delta_{bg}$ , and PapayaFormer, a hybrid architecture with calibrated evidential confidence that abstains on unseen conditions and runs in —ms on commodity hardware. Under the corrected protocol the state of the art is —, not 99%, and that is the number the field should be trying to improve.

Future work: (i) multi-label co-infection and severity regression; (ii) few-shot adaptation to new geographies with the low-data curve as the design target; (iii) on-device continual learning from farmer feedback with privacy guarantees; (iv) extension of the leakage audit to the wider PlantVillage-derived literature, where we expect the same correction applies; (v) a prospective field trial measuring agronomic outcome, not classification accuracy.

## DECLARATIONS

**CRedit authorship contribution statement.** *Mursalin Hawlader*: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Visualization. *Md. Shovon*: Conceptualization, Methodology, Validation, Supervision, Writing — review & editing. [Third Author]: Resources (field access, pathological annotation), Validation, Writing — review & editing.

**Declaration of competing interest.** The authors declare no known competing financial interests or personal relationships that could have appeared to influence this work.

**Funding.** This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. [Amend if applicable.]

**Data availability.** The curated field dataset (D6, NU-PapayaField), split manifests with per-image perceptual hashes and group assignments, expert lesion masks, and all evaluation scripts are available at [Zenodo DOI] and [GitHub URL] under CC BY 4.0 / MIT. Public sources D1–D5 are cited to their original DOIs and are not redistributed.

**Code availability.** [GitHub URL], tagged release matching this manuscript, with environment lockfile and a single command reproducing every table.

**Ethics.** No human or animal subjects. Field images were collected with landowner permission; no identifiable persons or property appear; GPS coordinates are coarsened to district level prior to release.

**Declaration of generative AI use.** During the preparation of this work the authors used [tool] for [language editing / code scaffolding / literature organisation]. The authors reviewed and edited the content and take full responsibility for the content of the publication. (Required by Elsevier and most IEEE venues — do not omit; omission is now a desk-reject trigger.)



## Acknowledgements. —

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- [13] [PAPAYA REF A] — *to be added from your screening pass (Table 1)*.
- [14] [PAPAYA REF B] — *to be added*.
- [15] [DATASET DOI D1] — *cite the Mendeley/Kaggle record exactly as the depositor specifies*.
- [16] [DATASET DOI D2] — *to be added*.
- [17] [SAM / segmentation reference] — *to be added*.
- [18] [DINOv2 reference] — *to be added*.

## Appendix A. REPRODUCIBILITY CHECKLIST

| #  | Item                                                             | Status | Where / artefact |
|----|------------------------------------------------------------------|--------|------------------|
| 1  | Dataset sources cited with persistent identifiers                | —      | Table 3          |
| 2  | Licences stated and compatible with redistribution               | —      | Table 3          |
| 3  | Class definitions with visual criteria                           | Y      | Table 4          |
| 4  | Inter-annotator agreement reported ( $\kappa$ )                  | —      | §3.3             |
| 5  | Duplicate/near-duplicate audit performed                         | Y      | Alg. 1           |
| 6  | Leakage rate $\rho$ reported per split                           | Y      | Table 6          |
| 7  | Split manifests released (image $\rightarrow$ fold, hash, group) | —      | repo / splits    |
| 8  | Random seeds fixed and listed                                    | Y      | Table 8          |
| 9  | Number of runs $\geq 5$ ; variance reported                      | Y      | §5.3             |
| 10 | Confidence intervals reported (method stated)                    | Y      | §5.3             |
| 11 | Significance tests with multiplicity correction                  | Y      | §5.3             |
| 12 | Effect sizes reported alongside $p$ -values                      | Y      | §5.3             |
| 13 | Hyperparameter search space and budget disclosed                 | Y      | App. Appendix B  |
| 14 | Test set used once; no test-set tuning                           | Y      | §8               |
| 15 | Baselines trained under identical protocol                       | Y      | §4.6             |
| 16 | Compute cost and carbon estimate reported                        | —      | Table 9          |
| 17 | Software versions pinned (lockfile)                              | —      | repo / env       |
| 18 | Model checkpoints released                                       | —      | Zenodo           |
| 19 | Single-command reproduction script                               | —      | make reproduce   |
| 20 | Deviations from pre-registered plan logged                       | —      | repo / PREREG.md |
| 21 | Failure cases shown in qualitative figures                       | Y      | Fig. 6           |
| 22 | Negative/null results reported                                   | —      | §6               |
| 23 | Generative-AI use disclosed                                      | Y      | Declarations     |

## Appendix B. HYPERPARAMETER SEARCH AND SENSITIVITY

| Hyperparameter                     | Search space                             | Selected (criterion: val macro- $F_1$ ) |
|------------------------------------|------------------------------------------|-----------------------------------------|
| Peak learning rate                 | $\{1, 3\} \times \{10^{-4}, 10^{-3}\}$   | —                                       |
| Weight decay                       | $\{0.01, 0.05, 0.1\}$                    | —                                       |
| $\lambda_1$ (class-balanced focal) | $\{0.5, 1.0, 2.0\}$                      | —                                       |
| $\lambda_2$ (consistency)          | $\{0, 0.1, 0.5, 1.0\}$                   | —                                       |
| $\lambda_3$ (attention reg.)       | $\{0, 0.001, 0.01, 0.1\}$                | —                                       |
| $T_{\text{anneal}}$ (KL anneal)    | $\{5, 10, 20, 40\}$                      | —                                       |
| MSLA dilation set                  | $\{(1, 2), (1, 2, 3), (1, 2, 3, 5)\}$    | —                                       |
| MSLA reduction $r$                 | $\{4, 8, 16\}$                           | —                                       |
| $\theta_e$ (embedding link)        | $\{0.90, 0.925, 0.95, 0.975\}$           | —                                       |
| Search strategy / budget           | Random search, — trials, val fold 1 only | —                                       |

**Documented failure modes to report honestly.** (i) Without KL annealing, the evidential head collapses to  $u \approx 1$  for all inputs within — epochs and accuracy stalls at chance — report the epoch and the curve. (ii) With  $\lambda_3$  too large, MSLA masks collapse to a single point and macro- $F_1$  drops by — pp. (iii) Strong hue jitter (excluded in Table 7) degrades  $c_2/c_6$  separation by — pp — include this as the empirical justification for the exclusion, since a reviewer will otherwise assume the exclusion was arbitrary.

## Appendix C. PER-CLASS RESULTS

| Class                     | Precision | Recall | $F_1$ | Support | AUROC | $F_1$ (S3 shift) | $F_1$ (INT8) | $\Delta$ INT8 |
|---------------------------|-----------|--------|-------|---------|-------|------------------|--------------|---------------|
| Healthy ( $c_1$ )         | —         | —      | —     | —       | —     | —                | —            | —             |
| PRSV ( $c_2$ )            | —         | —      | —     | —       | —     | —                | —            | —             |
| Anthrachnose ( $c_3$ )    | —         | —      | —     | —       | —     | —                | —            | —             |
| Powdery mildew ( $c_4$ )  | —         | —      | —     | —       | —     | —                | —            | —             |
| Bacterial spot ( $c_5$ )  | —         | —      | —     | —       | —     | —                | —            | —             |
| Mite/deficiency ( $c_6$ ) | —         | —      | —     | —       | —     | —                | —            | —             |
| <b>Macro avg.</b>         | —         | —      | —     | —       | —     | —                | —            | —             |

## Appendix D. DATASET METADATA SCHEMA

| Field                    | Type        | Description                                             |
|--------------------------|-------------|---------------------------------------------------------|
| image_id                 | string      | Stable unique identifier; filename-independent          |
| leaf_id                  | string      | Physical leaf identity — <b>the split key</b>           |
| plant_id / plot_id       | string      | Coarser grouping for hierarchical shift analysis        |
| district                 | string      | Coarsened location (no raw GPS in release)              |
| date, time               | ISO 8601    | Season and illumination proxy                           |
| device                   | string      | Phone model; a shift variable, not noise                |
| illumination_class       | enum        | {direct sun, overcast, shade, dusk}                     |
| phenostage               | enum        | {seedling, vegetative, flowering, fruiting}             |
| label                    | enum        | $c_1 \dots c_8$                                         |
| annotator_id, confidence | string, 1–5 | Per-annotator label and self-rated confidence           |
| adjudicated              | bool        | Whether a third rater resolved disagreement             |
| lab_confirmed            | bool        | Serological/PCR confirmation available                  |
| coinfection_flag         | bool        | Multiple syndromes present                              |
| md5, phash               | string      | Released so others can audit leakage against our splits |
| group_id                 | string      | Assigned by Alg. 1                                      |
| split_S2_fold            | int         | Released fold assignment for exact reproduction         |

## Appendix E. LITERATURE SCREENING RECORD

| Stage                                                                             | Count | Note                                                                 |
|-----------------------------------------------------------------------------------|-------|----------------------------------------------------------------------|
| Records identified (Scopus, WoS, IEEE Xplore, ScienceDirect, ACM, Google Scholar) | —     | Query string in §2; search date —                                    |
| Duplicates removed                                                                | —     |                                                                      |
| Title/abstract screened                                                           | —     |                                                                      |
| Excluded at screening                                                             | —     | Reasons tabulated                                                    |
| Full text assessed                                                                | —     |                                                                      |
| Excluded at full text                                                             | —     | Reasons tabulated (no papaya, no evaluation detail, not retrievable) |
| <b>Included in Table 1</b>                                                        | —     |                                                                      |

A PRISMA-style flow diagram is optional for a primary study but converts the related-work section from an opinion into a reproducible procedure. Include it if targeting a review-heavy venue.

## Appendix F. ANTICIPATED REVIEWER OBJECTIONS AND PREPARED RESPONSES

*Not for submission — an internal document. Every item below is an objection this paper will attract; each must be answerable from the manuscript itself before you submit.*

| #  | Likely objection                                                          | Where the manuscript answers it                                                                                                                                                                     |
|----|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | "Accuracy is lower than published work — the method is weaker."           | §7, Table 19: we report our own S1 number under the legacy protocol for direct comparability, and show the gap is protocol, not capability.                                                         |
| 2  | "The dataset is small."                                                   | RQ10 low-data curve; group-level bootstrap CIs; power analysis in §8. State $n$ at group level, not image level — that is the honest sample size.                                                   |
| 3  | "Novelty is limited; MSLA is a variant of CBAM."                          | Ablation Table 16 isolates MSLA; the framed contribution is the protocol (C1, C2), with the architecture secondary. <i>Do not oversell the module.</i>                                              |
| 4  | "Why not just use a large pretrained foundation model?"                   | Include a CLIP/DINOv2 linear-probe and a zero-shot VLM row in Table 10. If a frozen foundation model wins, report that — it is a finding, not a defeat.                                             |
| 5  | "Evidential deep learning is known to be miscalibrated in some settings." | Report ECE with and without EDL (Table 15) and compare against temperature-scaled softmax and deep ensembles; do not claim EDL is uniformly superior.                                               |
| 6  | "Leakage claim about prior work is unsubstantiated."                      | Never assert that a specific paper leaked. Assert that its <i>stated</i> split procedure cannot rule leakage out, quote the procedure verbatim, and demonstrate the effect on the same public data. |
| 7  | "Field dataset annotation quality is unclear."                            | $\kappa$ , adjudication rate, <code>lab_confirmed</code> fraction, annotator qualifications.                                                                                                        |
| 8  | "Edge numbers depend on the device."                                      | Two devices, both fully specified, median and p95, with the measurement script released.                                                                                                            |
| 9  | "No comparison with object detection / segmentation approaches."          | Scope statement in §1: classification of a leaf-centred image is the deployment modality; detection is future work. Justify, do not ignore.                                                         |
| 10 | "Statistical tests are missing / inappropriate."                          | §5.3 pre-specifies test, correction and effect size per family.                                                                                                                                     |
| 11 | "Results are not reproducible."                                           | App. Appendix A, released splits with hashes, <code>make reproduce</code> .                                                                                                                         |
| 12 | "Practical impact is speculative."                                        | §7 cost matrix and referral workload; prospective trial named as future work rather than implied.                                                                                                   |

## Appendix G. EXECUTION PLAN

*Not for submission — your working schedule. Ordered so that the paper’s central claim (RQ2) is testable in week 3; if it does not hold on your data, you learn that before investing in the architecture.*

| Week | Task                                                                                                             | Deliverable / gate                                                         |
|------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1    | Acquire D1–D5; verify licences and DOIs; build the unified loader and label mapping.                             | data/ + DATASETS.md with verified links                                    |
| 2    | Implement Alg. 1; compute $\rho$ for random splits on every source; calibrate $\theta_c$ on hand-labelled pairs. | <b>Gate: is <math>\rho</math> large?</b> If yes, the paper has its result. |
| 3    | Train ResNet-50 + EfficientNet-B0 under S1 and S2.                                                               | Table 11 filled — the headline                                             |
| 4    | Segmentation + Clever-Hans probe on all sources.                                                                 | Table 12 filled                                                            |
| 5–6  | Field collection (D6): $\geq 3$ districts, $\geq 3$ devices, dual annotation, $\kappa$ .                         | Dataset + metadata CSV                                                     |
| 7    | Expert lesion masks on the XAI subset.                                                                           | Masks for App. Appendix C/XAI                                              |
| 8    | Implement PapayaFormer; MSLA and evidential head unit-tested.                                                    | Passing tests, overfit-on-10-images sanity check                           |
| 9    | Full baseline sweep, 5 seeds $\times$ 5 folds.                                                                   | Table 10                                                                   |
| 10   | Ablations, corruption suite, cross-source, low-data.                                                             | Tables 13, 16, Figs. 4, 7                                                  |
| 11   | Open-set, calibration, selective prediction.                                                                     | Tables 14, 15                                                              |
| 12   | XAI evaluation incl. sanity checks.                                                                              | Table 17, Fig. 6                                                           |
| 13   | Edge export, quantisation, on-device measurement.                                                                | Table 18                                                                   |
| 14   | Statistics, CIs, corrections, all figures final.                                                                 | Camera-ready figures                                                       |
| 15   | Write Discussion/Conclusion; complete checklist; Zenodo deposit.                                                 | DOI minted                                                                 |
| 16   | Internal review against App. Appendix F; language pass; submit.                                                  | Submission                                                                 |

**Venue targets.** Verify current scope, impact factor and APC before choosing — these change annually.

| Venue                                                           | Type           | Fit note                                                                      |
|-----------------------------------------------------------------|----------------|-------------------------------------------------------------------------------|
| Computers and Electronics in Agriculture (Elsevier)             | Q1, APC/hybrid | Strongest fit: agricultural ML with field validation; expects agronomic depth |
| Smart Agricultural Technology (Elsevier)                        | Q1/Q2, OA      | Deployment- and edge-oriented; good fit for the RQ9 material                  |
| Frontiers in Plant Science — Technical Advances                 | Q1, OA         | Precedent set by [1]; receptive to the correction narrative                   |
| IEEE Access                                                     | Q1/Q2, OA      | Fast, tolerant of a broad benchmark contribution; lower prestige              |
| Scientific Reports (Nature)                                     | Q1, OA         | Methodology-correction framing travels well                                   |
| Plant Methods (BMC)                                             | Q1, OA         | Strong fit for the protocol/benchmark contribution specifically               |
| Ecological Informatics / Artificial Intelligence in Agriculture | Q1/Q2          | Reasonable alternates                                                         |

**Before you submit — non-negotiables.**

1. Every  $\dashv$  replaced by a number you produced, or the row deleted. Never estimate a result.
2. Every  $\dagger$  resolved against the primary source. Never cite a DOI you have not opened.
3. Every citation in the bibliography read by you, at least in abstract and method. Fabricated or mis-attributed citations end careers, and reviewers in this field do check.
4. The claim about prior work stated as "the reported procedure does not exclude leakage", never as "they leaked".
5. Data and code deposited and the link live *before* submission, not promised.